

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of B. Pharm. Examination
University Theory Examination September 2018
BPH2003 Pharmaceutical Engineering

Date: 19/09/2018, Wednesday **Time:** 10:00 a.m. to 01:00 p.m. **Maximum Marks:** 75

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION – I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. The fluid flow in which the fluid particles in one layer do not mix with the fluid particles in the other layer is called as _____.
[A] laminar flow
[B] turbulent flow
[C] layer flow
[D] offline flow
2. The viscosity of a liquid _____.
[A] increases with increase in liquid temperature
[B] decreases with increase in liquid temperature
[C] is not affected by the change in liquid temperature
[D] is unpredictable
3. Generally, thermal conductivity of liquids is _____.
[A] very high
[B] very small
[C] varies from very small to very high
[D] varies from very high to very small
4. What is the formula for Reynolds number (Re)?
[A] $(Re) = (D \ v \ \rho) / \mu$
[B] $(Re) = (D \ \rho \ \mu) / v$
[C] $(Re) = (D \ v \ \mu) / \rho$
[D] $(Re) = (D \ v \ \mu) / \mu$
5. As the rate of feed increases, the size reduction _____.
[A] increases
[B] remains constant
[C] equals
[D] decreases

6. A separation technique which uses fractionating column to separate different liquid components in a mixture with different boiling points by heating mixture is called _____.
[A] fractional distillation
[B] filtration
[C] evaporation
[D] distillation
7. Water can be separated from mud by help of _____.
[A] size separation techniques
[B] filtration
[C] crystallization
[D] flash distillation
8. Solid particles which are left on filter paper while filtration is called _____.
[A] residue
[B] filtrate
[C] solvent
[D] solute
9. Which filter consists of alternately arranged plates and frames where frames are fitted with filter cloth or filter pad?
[A] Plate & Frame Filter
[B] Pressure leaf filter
[C] Rotary vacuum filter
[D] Metafilter
10. Which centrifuge contains perforated bowls with a filter bag of nylon or cotton?
[A] Basket centrifuge
[B] Tubular-bowl centrifuge
[C] Multi chamber centrifuge
[D] Disc-bowl centrifuge
11. After critical moisture content _____ starts.
[A] saturated drying region
[B] unsaturated drying region
[C] constant drying region
[D] supersaturated drying region
12. The following is used to transport materials having flat bottoms.
[A] Belt conveyor
[B] Roller conveyor
[C] Chain conveyor
[D] Slip conveyor
13. Which of the following is a characteristic of a horizontal tube evaporator?
[A] Agitation is provided only by bubbles leaving the evaporator as vapor
[B] The tube bundle is arranged vertically, with the solution inside the tubes condensing outside
[C] To handle viscous solution a pump is used to force liquid upwards
[D] Also called short vertical tube evaporator
14. Which of the following is not physical hazard agent?
[A] Falls
[B] Electricity
[C] Inhalation
[D] Sulphuric acid

15. Which of the following colour is used for radiation hazard?
 [A] Red
 [B] Orange
 [C] Green
 [D] Purple
16. Class-A fire consists of fire due to _____.
 [A] wood
 [B] oil
 [C] transformer
 [D] chemical
17. Water is used to extinguish _____.
 [A] class-A fires
 [B] class-B fires
 [C] class-C fires
 [D] class-D fires
18. _____ is best suited to extinguishing oil or flammable liquid fire.
 [A] Soda acid
 [B] Vaporizing liquid
 [C] Foam
 [D] Dry chemical
19. The most important objective behind plant layout is _____.
 [A] Overall simplification and ease in integration of various functions
 [B] Economy in machines
 [C] Maximum travel time in plant
 [D] Minimum work-in-progress
20. A low unit cost can be achieved by following _____.
 [A] Process layout
 [B] Product layout
 [C] Fixed position layout
 [D] Functional layout

SECTION – II

- | | | |
|--|---|----|
| Q 2 | Enlist the objectives of material handling systems. Explain, with appropriate diagram, the different types of conveyors. | 10 |
| Q 3 Attempt any <u>TWO</u> of the following | | |
| A | Explain the principle, construction, working and uses of double cone blender along with the appropriate diagram. | 05 |
| B | Explain the principle, construction, working and uses of fluidized bed dryer along with the appropriate diagram. | 05 |
| C | Explain the principle, construction, working and uses of Perforated basket centrifuge along with the appropriate diagram. | 05 |

SECTION – III

- Q 4** Attempt any **ONE** of the following
- A** Explain the principle, construction, working, uses, advantages, disadvantages of cyclone separator along with appropriate diagram. **10**
- B** Define size reduction. What are the objectives of size reduction? Explain the mechanisms of size reduction with a suitable diagram. **10**
- Q 5** Attempt any **FIVE** of the following
- A** Describe Mier's theory for supersaturation with limitations. **05**
- B** Derive Bernoullis's equation with proper assumption. **05**
- C** Discuss various factors affecting on selection of materials for plant Construction. **05**
- D** Derive an equation for overall heat transfer coefficient. **05**
- E** Define corrosion. Describe galvanic corrosion and its prevention. **05**
- F** Explain principle and working of filter leaf. **05**
-